The Reggae Shark Protocol (v1.3.2): 42 Lean 4 Theorems, Pinned-Key Ed25519 Ledger Verification, 200-Bit Interval Patankar Bridge & Bandlimited LRSC

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Abstract

The Reggae Shark Protocol (v1.3.2) is a verifiable execution kernel and multi-cloud AI gateway that binds agent prompt dispatches to four verification layers: (1) an append-only, SHA-256 hash-chained and RFC 8032 Ed25519-signed **IntentSeal Verifiable Research Kernel (VRK v0.3.1)** ledger requiring an explicit private seed or child-read key file, mandatory pinned `expected_pubkey` verification, append-only `fsync` persistence, timestamped signed head commitments (`intentseal-vrk-head-v2` with `issued_at`) and minimum-length export verification (`verify_exported_ledger`), an N-way associative Merkle-DAG join-semilattice, and stateful child-process key isolation (`IsolatedSignerSubprocess` enforcing `prev_hash == last_signed_digest`); (1B) a thread-synchronized **Request-Lifecycle Typestate and Worker-Epoch Isolation** state machine (`TypestateLifecycle.lean`, 16 zero-`sorry` theorems including unique reachability of `SealedActive` from `Uninit`) verified by differential trace testing and CI `#eval` diffing against `data/lean_transition_table.json`; (2) a per-tenant **4D Soft-Wall ODE Governor** $(\gamma, H_{\text{obs}}, \dot{I}, \Psi)$ backed by 17 4D boundary theorems (`ExistenceVerification.lean`), 9 3D soft-wall theorems (`BoundaryVerification.lean`), runtime `float64` box checks (`inside_B4_checked`), and **200-bit outward-rounded `mpmath.iv` Patankar–Euler interval enclosures** (`inside_B4_mp_iv_certified`), plus an illustrative 120-segment Duhamel parameter study on `data/wp19_segments.json`; and (3) a **Bandlimited LRSC** operator ($M = 99, K = 11$) with linear perturbation budget scaling (`perturbation_budget_scaling`).

1 Four-Layer Verification Architecture & Provenance (42 Lean 4 Theorems)

Every request submitted to `MultiProviderGateway.process_request` or the local loopback HTTP proxy (`POST http://127.0.0.1:8000/v1/execute`) traverses four deterministic verification gates:

Layer	Module / Artifact	Enforced Mathematical & Cryptographic Invariant

Layer 1	`shark_protocol/kernel.py`
	`SECURITY.md`
	`data/release_head.json`
		Pinned-Key Ed25519 VRK v0.3.1 Ledger, Timestamped Head & Stateful Child Signer: Anchored at trust root `f010a99d...05a0d0f` (10.5281/zenodo.23018459); requires `cryptography` and explicit seed/key file, requires `expected_pubkey` in `verify_chain_list`, enforces `issued_at` and `min_length` in `verify_exported_ledger`, and enforces `prev_hash == last_signed_digest` in `IsolatedSignerSubprocess`.

Layer 1B	`lean/TypestateLifecycle.lean` `data/lean_transition_table.json`	**Typestate, Epoch & Differential Trace Suite (16 Theorems) + RLock:** 6-state request automaton (`UNINIT` $\to \cdots \to$ `ZEROED_FREED`) with absorbing `ZEROED_FREED`, unique reachability (`sealed_active_unique_trace_from_uninit`), and CI diffing of Lean `#eval` JSON against `data/lean_transition_table.json`.
Layer 2	`lean/ExistenceVerification.lean` `lean/BoundaryVerification.lean`	**4D Soft-Wall ODE Governor ($\mathcal{B}_4$, 26 Theorems) & 200-Bit Interval Bridge:** Adaptive RK4, `float64` Patankar–Euler check (`inside_B4_checked`), and 200-bit `mpmath.iv` interval enclosure (`step_ode_patankar_mp_interval`, `inside_B4_mp_iv_certified`) inside $\mathcal{B}_4 = [0,10] \times [0,100] \times [0, \dot{I}_{\max,4D}] \times [0,1]$.
Layer 3	`LRSCCompressor` (`kernel.py`)	**Bandlimited LRSC Odd-Ring Projection (11/99 = 1/9):** Projects context onto the $M = 99$ odd-ring filter V_{ring} ($\sigma_{\min}(S_{11}) = 0.181646 > 0$, 10.17605/OSF.IO/NM5BW), bounds perturbation scaling under assumed $L \geq 0$ ([6]), and binds distinct compressed/uncompressed SHA-256 digests.
Layer 4	`shark_protocol/gateway.py`	**Token-Authenticated Multi-Tenant Gateway & `ThreadingHTTPServer`:** Authenticates Bearer tokens via `sha256(token)` $\to$ `tenant_id` (HTTP 401 on unknown token), persists per-tenant ledgers on disk, returns HTTP 503 at `max_tenants=128` without eviction, enforces rate limits (HTTP 429), HTTP 413 caps, and a 10 s socket timeout.

Table 1: Four-layer verification architecture of The Reggae Shark Protocol (v1.3.2).

Canonical Lean 4 Module Provenance & CI Axiom Gate (42 Zero-sorry Theorems): Specifically, `lean/BoundaryVerification.lean` (SHA-256: `49229b61...e417`, 9 theorems) is byte-for-byte identical to Physics v1.5 Appendix A (10.5281/zenodo.23534449), and `lean/ExistenceVerification.lean` (SHA-256: `eec94868...bcae`, 17 theorems) is byte-for-byte identical to *The Theory of Bounded Existence* v1.0.0 (10.5281/zenodo.23608244). Together with the 16 theorems in `lean/TypestateLifecycle.lean`, `lean/AxiomsCheck.lean` and `tools/check_lean_axioms.py` verify in CI that all **42 theorems** depend only on subsets of {`propext`, `Classical.choice`, `Quot.sound`} with zero `sorryAx`, and that Lean's `#eval dumpTransitionTableJson` output matches `data/lean_transition_table.json`.

2 What Is and Is Not Modelled (Epistemic Claim Firewall & Threat Model)

Epistemic Standard (Build. Attack. Falsify. Preserve.): Every theorem and runtime guard in this repository is bounded by explicit domain assumptions, `SECURITY.md` threat boundaries, and `scope.does_not_imply` declarations.

- **Application-Layer Typestate & Stateful Subprocess Signer vs. OS Ring-0:** `RequestSlot` and `WorkerEpochGuard` enforce typestate invariants in Python, while `IsolatedSignerSubprocess` loads its key from a child-read `seed_file` (or generates an ephemeral key in-child) and refuses to sign any entry with `prev_hash` $\neq$ `last_signed_digest`. If the parent gateway is compromised mid-session, it can still request signatures on new forward entries extending `last_signed_digest`, but cannot rewrite history, fork an old digest, sign a truncated head, or extract the raw key. Theorem 14 (`projection_independent_of_secret`) is a definitional projection lemma; process isolation rests on the runtime child-process design (`test_32`, `test_41`).
- **Real Arithmetic in Lean vs. 200-Bit `mpmath.iv` & IEEE-754 Runtime Checks:** Lean 4 Theorems 6, 7, 9, 10, 13, and 15 are proved over $\mathbb{R}$; `step_ode_patankar_interval` performs a fast `float64` 2-ULP check (`inside_B4_checked`), whereas `step_ode_patankar_mp_interval` bridges discrete steps to $\mathbb{R}$ via 200-bit outward-rounded `mpmath.iv` intervals (`inside_B4_mp_iv_certified`, `INTERVAL_CERTIFIED_RESULT`).
- **Illustrative WP19 Parameter Study vs. Continuum Navier–Stokes:** `evaluate_wp19_piecewise_duhamel_study` evaluates the 120-segment Duhamel recurrence on the assumed segment rates loaded from `data/wp19_segments.json` (SHA-256: `a78a67c8...99319`, 10.5281/zenodo.23034545, `NUMERICAL_OBSERVATION`); in-repo hash pinning catches accidental file corruption, while external tamper resistance is anchored by the release notes and Zenodo record. Continuum 3D Navier–Stokes regularity remains `OPEN`.

3 Layer 2 Equations & Verification Suite (42/42 Tests Passing)

The Layer 2 governor evolves $x(t) = (\gamma(t), H_{\text{obs}}(t), \dot{I}(t), \Psi(t))$ under the smooth (C^∞) soft-wall system [3]:

$$\frac{d\gamma}{dt} = a\dot{I}\mathcal{E}_{\text{avail}}\left(1 - \frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \qquad \frac{dH_{\text{obs}}}{dt} = c\gamma\,(H_{\max} - H_{\text{obs}}) - \dot{I}_{\text{erase}}\tanh\left(\frac{H_{\text{obs}}}{\epsilon_H}\right), \tag{1}$$

$$\frac{d\dot{I}}{dt} = e\tanh\left(\frac{\dot{I}}{\epsilon_I}\right)\frac{dH_{\text{obs}}}{dt} + f\mathcal{P}_{\text{eff}}(\gamma,\Psi) - g\dot{I}, \qquad \frac{d\Psi}{dt} = \kappa_{\text{meta}}R_{\text{self}}(x)\,(1-\Psi) - \mu_\Psi\Psi, \tag{2}$$

where $\mathcal{P}_{\text{eff}}(\gamma,\Psi) = \mathcal{P}_0 + \lambda_{\text{sim}}r_{\text{LRSC}}\gamma + \lambda_{\text{lucid}}\Psi$, $R_{\text{self}}(x) = \text{fb}(x)^2/(\text{fb}(x)^2 + (f\mathcal{P}_0)^2 + \sigma_0^2)$, and $\text{fb}(x) = e\tanh(\dot{I}/\epsilon_I)(dH_{\text{obs}}/dt)$.

1. **Patankar–Euler Real-to-Float Bridge (200-Bit `mpmath.iv` Certification vs. `float64` Check):** Updates each coordinate via

$$x_{k+1} = \frac{x_k + \Delta t\,P(x_k)\,x_{\max}}{1 + \Delta t\,(P(x_k) + D(x_k))}, \qquad P(x_k) \geq 0,\ D(x_k) \geq 0,\ 1 + \Delta t(P+D) \geq 1, \tag{3}$$

   paired with Python `float64` floor `FLOAT64_NORMAL_MIN` $= 2.225 \times 10^{-308}$ (`inside_B4_checked`) and `step_ode_patankar_mp_interval` (200-bit outward-rounded `mpmath.iv` intervals of width $< 10^{-45}$ agreeing with `float64` within $< 10^{-12}$, `inside_B4_mp_iv_certified`, Theorem 13 `patankar_interval_rounding_in_box`, `test_15`, `test_21`, `test_31`).

2. **Pointwise Decay Bound, Token Registry Authentication, Per-Tenant Persistence & Non-Evicting Cap:** Theorem `psi_decay_floor_bound` proves $d\Psi/dt \geq -\mu_\Psi\Psi$ (yielding $\Psi(t+\Delta t) \geq \Psi(t)e^{-\mu_\Psi\Delta t}$ via Grönwall), while `MultiProviderGateway` authenticates Bearer tokens via `sha256(token)` $\rightarrow$ `tenant_id` (HTTP 401 on

unknown token), persists per-tenant ledgers to `<ledger_dir>/<tenant_id>.jsonl`, returns HTTP 503 at `max_tenants=128` without eviction, rate-limits requests (HTTP 429), and clamps $\mathcal{P}_0 \geq 1.0 \gg 10/99$ (`test_16`, `test_22`, `test_29`, `test_38`, `test_40`).

3. **Prompt-Committed Bandlimited LRSC Witness & Perturbation Budget Scaling:** `LRSCCompressor.compress` injects a slot-0 anchor to prevent scalar-multiple collisions (`test_27`), projects onto V_{ring} ($\sigma_{\min}(S_{11}) = 0.18164624 > 0$ [4]), evaluates linear perturbation budget scaling $L_{\text{local}}\eta_{\perp}(\tilde{x})$ on $\|\tilde{x}\|_2 \leq 1$ [6] (Theorem 10 `perturbation_budget_scaling`), and is verified against empirical softmax attention perturbations (`test_17`, `test_23`, `test_27`, `test_38`).

4. **Inductive Typestate Trace Safety, Unique Reachability & Differential Equivalence:** Theorems 1–5 and 16 prove over arbitrary event lists `evs : List LifecycleEvent` that `ZeroedFreed` is strictly absorbing, {`AbortedPoisoned`, `ZeroedFreed`} is closed, and `runTrace Uninit evs = SealedActive` $\leftrightarrow$ `[BeginInit, SealSuccess]` (`sealed_active_unique_trace_from_uninit`), while `test_36` and `test_42` verify equivalence against `data/lean_transition_table.json`.

5. **Pinned-Key Ed25519 Signing, Timestamped Head Anti-Replay, Stateful Child Signer & Merkle-DAG:** `verify_exported_ledger` requires a timestamped `signed_head` (`intentseal-vrk-head-v2` with `issued_at`) and `min_length` (reference head hash in `data/release_head.json`: `866e1919b5a3e9e1...6608ee5b`, `test_35`, `test_39`), `IsolatedSignerSubprocess` loads its key from `seed_file` and enforces `prev_hash == last_signed_digest` (`test_32`, `test_41`), and `merge_divergent_ledgers` satisfies $(A \sqcup B) \sqcup C = A \sqcup (B \sqcup C)$ (Theorem 12, `test_19`, `test_25`).

6. **Illustrative 120-Segment Piecewise Duhamel Parameter Study (`data/wp19_segments.json`):** Theorem 15 (`duhamel_piecewise_contraction_bound`) proves $\alpha(\alpha e_0 + d_1) + d_2 < G(Ge_0 + d_1) + d_2$ for $0 \leq \alpha < G$ and $d_1 > 0$, and `evaluate_wp19_piecewise_duhamel_study` evaluates the 120-segment recurrence in 200-bit `mpmath.iv` arithmetic (10.5281/zenodo.23034545, `test_33`, `test_34`).

References

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